INTEGRATING LEAN AND GREEN PRINCIPLES TO REDUCE MATERIAL WASTE IN CONSTRUCTION AND DEMOLITION: A LITERATURE REVIEW

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EXECUTIVE SUMMARY

The impact of using virgin materials and not being able to regenerate them has created the growing problem of a continuous depletion of natural finite resources; this issue has prompted a research focus on minimizing the waste of materials. This paper presents a review of a collection of literature into applications of lean and six-sigma methodology, concepts, or tools applied to address The Third Green Waste: Materials, with a specific focus on construction and demolition (C&D) waste. With C&D waste accounting for more than twice the total amount of generated municipal solid waste in the United States at 534 million tons, it has become a heavily researched and influential topic [1]. *Tools such as Value Stream Mapping, 5S implementation, 3R Hierarchy, Life Cycle Assessment, and The Last Planner System are discussed in depth. Results of implementing these lean tools provide promising results of reducing C&D waste; summarized findings are presented. Future research aims towards more sustainable reduction such as legislature and centralized standards, as well as education about waste sorting and recycling and methods which focus on reduction rather reuse or recycling.*

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1.0 INTRODUCTION

With a continuous depletion of natural finite resources, and lack of sustainable consumption, material waste is an ongoing problem creating vast environmental and financial issues. The impact of using virgin materials and not being able to reuse them has caught up to us as landfills now produce 23% of all methane generated by human-related sources[1]; The Environment Protection Agency concluded that more than 534 million tons of construction and demolition debris goes to landfills every year, while other sources of municipal solid waste contributed 258 million tons[1]. The environmental consequences of this waste through greenhouse gas emissions and resource depletion are astounding. Based on these findings, it can be concluded that material waste management is absolutely necessary to mitigate this growing issue.

While everyone must do his or her part in minimizing material waste, the construction and demolition industry can have a large and effective impact in material waste reduction. With this realization, there has been recent research focused on methods of material waste management in the construction and demolition industry. Typically, approaches of waste management are deemed as lean manufacturing. The ultimate goal through implementation of lean manufacturing principles is to eliminate waste through decreasing non-value adding activities, also known as waste. However, when the perspective of concern changes from the customer to the environment, this is referred to as Green Manufacturing. Utilizing knowledge of both lean and green manufacturing techniques and tools is an effective way to increase the overall efficiency and performance of a system, while also working to decrease negative effects on the

environment. With such promising outcomes, a combination of lean and green manufacturing has been applied to research approaches in order to aid in material waste management within the construction industry.

Lean Construction is a combination of lean manufacturing principles and practical development in design and construction used to reduce waste and work towards continuous improvement. However, lean construction is not only limited to the phase in which construction actually takes place; this practice is meant to be implemented along the entire process and is used by owners, architects, designers, engineering, constructor, suppliers, and end users. With lean tools such as Value Stream Mapping, 5S, Kanban, Life Cycle Assessment, Last Planner System, etc., the reduction of C&D waste is prioritized which is beneficial for both the environment and the project at hand.

2.0 METHODOLOGY

Papers ranging from 1994-2018 have been collected on the topic of material waste and material waste management in the construction and demolition industry through the application of lean and green techniques. Beginning with background research on sources and consequences of the waste of materials, an overview of the underlying problem will be described. Further, a deeper look is taken into methods used for C&D waste management. Finally, several case studies and research papers are discussed in which lean and green manufacturing tools and techniques are utilized to manage the waste of materials in the construction and demolition industry.

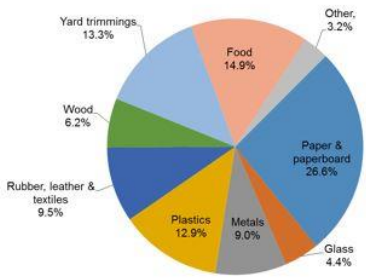
3.0 RESEARCH MOTIVATION

Ultimately, the waste of materials boils down to the negative environmental and financial impacts of pulling virgin materials and not being able to reuse them [2]. From the packet of paper printed one-sided, to the excess material taken off a block of aluminum to manufacture a part, material waste is occurring faster than the earth can replenish; in fact, in many cases, these materials are not replenishable. This creates a catastrophic situation where the material waste is piling up and resources are dwindling to replace them. This solid waste comes from a number of sources which can be broken down into 8 categories: Residential, industrial, commercial, institutional, construction and demolition, municipal services, processes, and agriculture, as shown below in Table 1 [3].

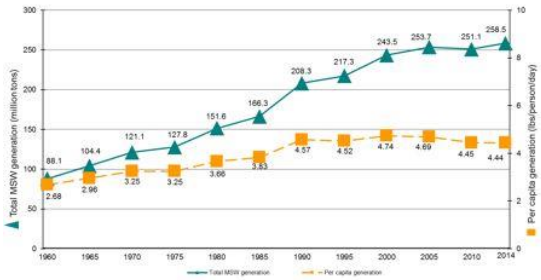
Table 1. Sources and Types of Solid Wastes [3]

Source	Typical waste generators	Types of solid wastes
Residential	Single and multifamily dwellings	Food wastes, paper, cardboard, plastics, textiles, leather, yard wastes, wood, glass, metals, ashes, special wastes (e.g., bulky items, consumer electronics, white goods, batteries, oil, tires), and household hazardous wastes.).
Industrial	Light and heavy manufacturing, fabrication, construction sites, power and chemical plants.	Housekeeping wastes, packaging, food wastes, construction and demolition materials, hazardous wastes, ashes, special wastes.
Commercial	Stores, hotels, restaurants, markets, office buildings, etc.	Paper, cardboard, plastics, wood, food wastes, glass, metals, special wastes, hazardous wastes.
Institutional	Schools, hospitals, prisons, government centers.	Same as commercial.
Construction and demolition	New construction sites, road repair, renovation sites, demolition of buildings	Wood, steel, concrete, dirt, etc.
Municipal services	Street cleaning, landscaping, parks, beaches, other recreational areas, water and wastewater treatment plants.	Street sweepings; landscape and tree trimmings; general wastes from parks, beaches, and other recreational areas; sludge.
Process (manufacturing, etc.)	Heavy and light manufacturing, refineries, chemical plants, power plants, mineral extraction and processing.	Industrial process wastes, scrap materials, off-specification products, slay, tailings.
Agriculture	Crops, orchards, vineyards, dairies, feedlots, farms.	Spoiled food wastes, agricultural wastes, hazardous wastes (e.g., pesticides).

Each year the Environmental Protection Agency produces a fact sheet titled Advancing Sustainable Materials Management; this report includes data on municipal solid waste (MSW) generation, recycling, combustion with energy recovery and landfilling from these sources [1]. Figure 1 below shows a pie chart generated from the total MSW generation by material as well as the trend in MSW generation rates.



Total MSW Generation (by material), 2014
258 Million Tons (before recycling, composting, or combustion with energy recovery)



MSW Generation Rates 1960 to 2014

Figure 1. Total MSW Generation in 2014 and Trends [1]

The rate at which waste generation is increasing is alarming to say the least and the total MSW generation accumulated to 258 million tons. However, this report does not include one of the aforementioned sources: *Construction and demolition*. The reason the EPA conducts studies separate from other MSW, is that construction and demolition produce more than twice the total amount of generated MSW at 534 million tons and has been

deemed a 'significant waste stream in the United States' [1]. This incredible amount of generated waste from a single source has been a driving factor for many studies across the world with a focus on methods of waste management for this industry. This paper will focus on these studies aiming to reduce material waste from the construction and demolition industry, with a focus on articles that utilized lean and green tools.

4.0 CONSTRUCTION AND DEMOLITION WASTE

Construction and demolition (C&D) waste is defined as the waste produced during new construction, renovation, and demolition of buildings and structures [4]. Materials included in C&D debris generation are steel, wood products, drywall and plaster, brick, clay tile, asphalt shingles, concrete and asphalt concrete and represents amounts from construction, renovation and demolition activities [1]. The composition of C&D waste generation by material is summarized below in Figure 2.

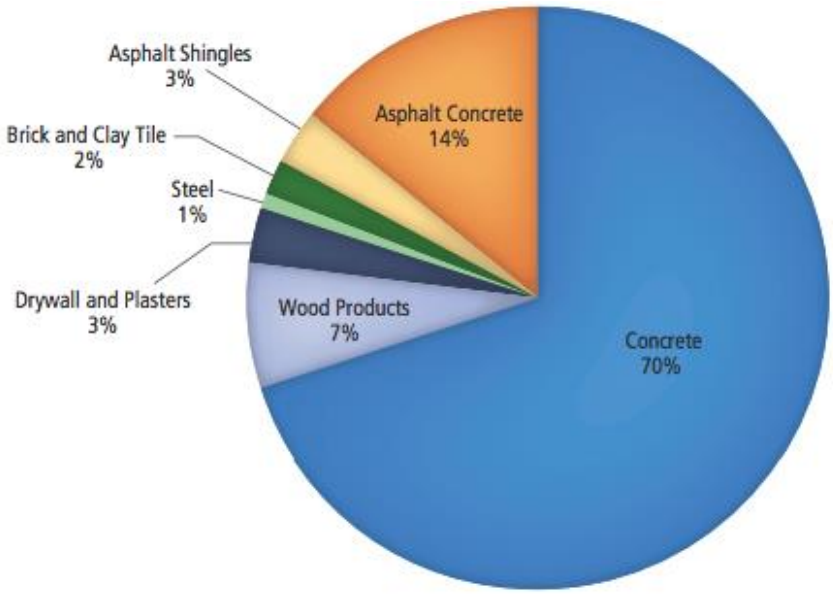


Figure 2. C&D Generation Composition by Material, 2014, 534 Million Tons (before recycling) [1]

While Figure 2 displays only the C&D waste generation within the United States, this issue is not confined to just one country. While over half of the volume of materials in landfills in the US is C&D waste, data available from several other countries around the world present similar findings [1]. A 2015 study found that C&D waste in China accounts for 30%-40% of the total municipal waste [5]. In a review of the waste streams in Europe, the total quantities of C&D waste in selected countries found that Germany, Austria, Italy, and Spain all had alarming quantities at 50, 25, 20, and 23 million tons, respectively [6]. With such large contributions to landfills all over the world, the construction and demolition industry has a lot of room for improvement and even a small reduction in C&D waste could result in a large improvement in overall material waste.

The cause of this astounding accumulation of C&D waste cannot be attributed to just one source. In a recent study, Polat et al. [7] aimed to identify the importance levels of the root causes of material waste during construction in the Turkish construction industry. Following an extensive literature review, 34 causes were identified and used to carry out a survey, which results would then be used to deduct reliability and ranking analyses to find which factors have the highest impact. The conclusion of this study established that frequent design changes and change orders, design and construction detail errors, and waste from cutting uneconomical shapes, had the highest level of impact on C&D waste generation [7]. Further, a study in The United Arab Emirates examined four construction projects and conducted an extensive survey which reduced causes of material waste to lack of awareness, excessive off-cuts resulting from poor design, and rework and variations [8]. In a similar fashion, Ekanayake et al. aimed to identify the most important

sources of waste in construction in Singapore where construction is recognized as a major contributor to waste in landfill sites; the study found that design changes while construction works are in progress was the most significant cause of waste generation on site [9]. Conclusions made from several other studies found agreeing causes mainly resulting from design issues, lack of attention paid to the sizes of the used products, lack of influence of contractors, and lack of knowledge about construction during design activities [10]–[12].

Unfortunately, realizing the sources of C&D waste is only half the battle. Both waste prevention, as well as management techniques, are necessary to mitigate this growing issue. To aid in this effort, research presenting lean and green tools to minimize C&D waste has been a growing topic of interest.

5.0 APPLYING LEAN AND GREEN TECHNIQUES TO MINIMIZE WASTE

Waste in the construction industry plays an important role for both efficiency as well as environmental impact. With such consequences, it only makes sense to utilize techniques that would optimize efficiency as well as decrease contribution to material waste and therefore lessen any harmful impacts on the environment. Hence, a combination of both lean and green thinking would be the ideal approach to waste management within this industry.

5.1 3R HIERARCHY

There is a general finding that construction and demolition waste management practices are guided by the 3R hierarchy- reduce, reuse, and recycle [13]. Yeheyis et al. noted that it is evident that over 75% C&D waste generated by the Canadian construction industry has a residual value and therefore could be recycled, salvaged and/or reused [14]. According to an extensive 2018 literature review, Ghisellini et al. concluded that recycling dominated as the most investigated principle at 69% of the studies, followed by reuse principles appearing in 17% of the literature, and reduction techniques in 3%; remaining papers utilized refurbish techniques as well as combinations of principles [15]. While this is not surprising, as recycling is often the method requiring the least effort of implementation, it is not the most effective distribution of methods. According to the hierarchy, the environmental actions rank as reduce, reuse, and finally recycle, based on greatest positive impact. With that being said, the ideal scenario would have reduction as top priority, followed by reuse and then recycling as a final resort. Huang et al. explored this principle in a C&D waste management analysis in China; results revealed that primary barriers of waste management included lack of building design standard, low cost for disposal, and an underdeveloped market for reused C&D waste [16]. Calvo et al. developed a dynamic model for C&D waste management in Spain revolving around driving policies based on incentives and tax penalties to encourage 3R implementation on site [17]. The simulation model used to assess the potential impact of the two policies concluded that a 30% reduction in C&D waste could be achieved in 12 years or less [17]. Several studies have established similar conclusions that introduction of the 3R hierarchy

to C&D waste management through policies, management implementation, incentives, etc. are greatly beneficial in reducing C&D waste, as well as increasing profitability to projects which utilize re-use of materials [18]–[22].

5.2 SUSTAINABLE VALUE STREAM MAPPING

Value Stream Mapping (VSM) is a lean tool used to help visualize all steps in a process while simultaneously identifying value added actions and non-value added actions. Beginning with a Current State Map, actions are taken to minimize non-value adding actions, also referred to as waste. Following the development of the Current State Map and extensive brainstorming, a Future State Map is created. In the application of using *Sustainable* Value Stream Mapping, the goal would be to focus on minimizing waste which negatively effects the environment; in this case, material waste such as concrete, wood, brick, etc. during the processes of construction and demolition. With the aforementioned identified sources of C&D waste, VSM can be extremely beneficial not only in the actual construction and demolition phases, but also in the design phase, which can act as a preventative measure to reduce C&D, waste. Matt et al. developed methodology which demonstrated how to design an integrated and customized value stream map for construction industry [23]. However, the authors note that only a limited selection of process patters were developed and further research is required to update these process patterns to strengthen the conceptual phase during the actual construction of buildings [23]. In another VSM approach, Yu et al. analyzes the home building industry and developed a Future State Map which included synchronized first-in, first-out lane-based flow, production leveling at pacemaker, and work

restructuring; while the main purpose of this model was to decrease lengthy delivery time and increase process reliability, material waste reduction is a bi-product of these implementations due to a more efficient overall process [24]. Further, a study on lean construction looked into the reduction and elimination of waste along the development phase of construction with the final step of the journey being the application of Value Stream Mapping to make all steps visible in the process and organization within the pre-project phase; Aziz concluded that organizational performance was improved and a more regulated process was produced [25]. In a similar approach, Arbulu et al. used VSM to analyze the waste of a re-engineered construction supply chain which had numerous inefficiencies existing mainly at interfaces between processes, disciplines or organizations. These inefficiencies are similar to the causes of C&D waste generation found in related literature reviews. The Future State Map developed by Arbulu et al. illustrated process improvements through the application of various supply chain management tactics and suggested implementation of the system across other construction supply chains as well to limit waste [26]. Another case study in a Brazilian construction company used Value Stream Mapping to identify the main problems and process wastes throughout the value flow and propose actions for improvement [27].

5.3 5S IMPLEMENTATION

5S is a commonly used lean tool, which stands for Sort, Set in Order, Shine, Standardize, and Sustain. While these five aspects are meant to work in congruency for full effect, sorting and standardization have been the main focus

within research of C&D waste management. Sorting of C&D waste can have a tremendous benefit on increasing the amount of waste, which is recycled or reused. For example, Poon et al. discussed how Hong Kong has adopted a strategy of depositing inert waste such as sand, bricks, and concrete into public filling areas for land reclamation and non-inert materials such as plastic, paper, and wood into waste landfills [28]. A survey was conducted to evaluate three waste sorting methods and results indicated that source separation has numerous advantages including less effort and better segregation of inert and non-inert wastes when compared to with central sorting. However, it was found that only through contractual requirements or legislation, on-site waste sorting is effectively implemented [28]. This brings up two important aspects of the 5S system-standardization and sustainability. Standardization is essential to sustain good practice, and sustaining the practice ensures that all previous steps carry out long-term positive effects. In the construction and demolition industry, standardization goes hand in hand with legislation and policy change; through these implementations, sustainability becomes a more likely possibility. For example, in Malaysia, a new standardization procedure of solid waste management was implemented and found to enhance social, economic and environmental efficiency as well as help resolve the dual crisis of nonrenewable resources depletion and environmental degradation [29]. Standard practices such as these would help tremendously in the management of C&D waste, and subsequently work towards a sustainably approach as government issued standards would be required by all sites to follow. Overall, apply 5S principles to construction and demolition waste

has proven to be extremely beneficial in not only management, but also reduction and reuse of C&D waste [30].

5.4 LIFE CYCLE ASSESSMENT

Life cycle assessment (LCA) is a widespread tool used to assess the environmental benefits and burdens associated to waste management systems and to identify strategies that will improve their performance [31]. This approach can be used in a multitude of applications towards sustainability and C&D waste management. For example, Yeheyis et al. presented a C&D waste sustainability index based on life cycle analysis which can be used to make decisions related to selection of material, sorting, recycle/reuse, and treatment or disposal options for C&D waste. Another LCA methodology was used to evaluate the environmental performance of the current regional management of C&D waste and to identify critical aspects and possible improving actions in a 2018 study presented by Borghi et al.; based on the results obtained from the LCA, the best-case scenario was for all C&D waste to be sent to recycling, which resulted in a significant decrease in ozone depletion effects as well as natural resources consumption effects [32]. The LCA approach was also taken by Lockrey et al. to evaluate recycling performance of concrete, a key component of construction and demolition waste in Vietnam [33]. Suggested improvements include construction demolition waste be used for more permanent applications such as building foundations and in new building materials; this approach confirmed benefits to technological advancements in concrete recycling as well as environmental benefits for all impact categories including greenhouse gas emissions, solid waste,

eutrophication and photochemical oxidation [33]. A similar study on LCA toward eco-efficiency concrete waste management in Malaysia was presented by Mah et al. and found that the third of four scenarios in which recycling scenarios showed at least 37.6% saving while the highest emission saving was 47% [34]. Life cycle assessment applied to C&D waste management is a growing topic of research and a valuable tool for assessing environmental impacts associated with all stages of a product's life from raw material to reuse, recycling, or disposal [35].

5.5 LAST PLANNER SYSTEM

The Last Planner System (LPS) is a production planning system designed to produce predictable work flow and rapid learning in programming, design, construction and commissioning of projects. Implementation of this program involves knowledge of Master Scheduling, Pull Systems, and Make Ready Work Plans. A study on LPS conducted by Mossman et al. describes many elements of the Last Planner process that eliminate waste as a by-product [36]. For example, the production plan and daily reviews involved reduce rework in the design, which is a large contributor to unnecessary C&D material waste [36]. Another application of LPS was considered in Nigeria to improve performance. Daniel et al. concluded that the Last Planner System provided many benefits including a reduction in design changes and design error [37]. A comparative study in which the Last Planner System was analyzed against the traditional construction process determined that problems with traditional construction management include no standard procedure, unsynchronized processes, and design changes; these issues result in significant material waste [38]. Munje et al. found through this

comparison that the implementation of LPS significantly reduced this wastage [38]. It is evident that Last Planner System implementation can prove to be extremely beneficial not only to reduce C&D waste, but also improve the efficiency of planning and control, and effectively improve the construction management performance [39].

6.0 CONCLUSIONS AND FUTURE RESEARCH AGENDA

In conclusion, while there are many efforts in place to reduce C&D waste, the implementation of lean and six sigma principles is a very promising and growing area of research. According to Small et al., implementation of lean construction is becoming more common with many examples of integrated project delivery success [40]. Lean construction is defined as the adoption of the lean mindset involving the focus of waste reduction and management of process flow [41]. Through implementation of both lean and green tools such as 3R Hierarchy, Value Stream Mapping, 5S, and Life Cycle assessment, C&D waste may be further reduced and managed. Furthermore, it is evident that the key determinates of lean construction implementation are the knowledge of lean construction, organizational structure and culture, and market factors [42]. These key determinates are the guiding factors toward the future research agent, in combination with recommendations set forth by regional government agencies [1].

The research agenda of C&D waste management leads towards a more in-depth look into actual implementation of lean and six sigma techniques into real world construction supply chains. While there are a multitude of simulation and modeling applications, which test scenarios of techniques being used to reduce C&D waste. In addition, the

recycling industry needs to be improved through increased professionalization, improved product standards, market development and better operating standards [43]. In addition, incentivizing recycling may prove to be a promising method of increasing the rate of recycling construction and demolition waste. As Huang et al. concluded, with increasing attention being put on sustainability issues, there is a trend for the national and local governments to enforce new CDW policies, regulations, and incentives [16]. Moreover, increasing the accessibility of recycling facilities is an important factor to increase recycling behavior, as is simple and more convenient methods of separating and storing recyclable waste. Utilizing the lean tool of 5S, the use of different containers with different identifying colors located in residential areas for source separation to minimize waste and maximizing recycling is recommended by Zhang et al. [43]. The authors further recommend education about waste sorting and recycling is necessary to promote the proper follow-through of these applications [43].

Through a collective effort of all personnel, and utilizing lean tools and principles to minimize waste across all processes, reduction in construction and demolition material waste is within reach. Each individual must take an active role to aim towards continuous improvement so the immense contribution of C&D waste to landfills may be reduced. With continued research and efforts, it is apparent that positive changes are being made and there is progress towards a cleaner Earth.

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